C769 Task 1 IT Capstone Topic Approval Form

The purpose of this document is to help you clearly explain your capstone topic, project scope, and timeline and to ensure that they align with your degree emphasis. Without clearly addressing each of these areas, you will not have a complete and realistic overview of your project, and your instructor cannot accurately assess whether your project will be viable for the purposes of these courses.

Complete this form and send it (via UGCapstoneIT@WGU.edu) to your instructor for approval. Once approved, you will receive a signed document in PDF format that you can upload as part of Task 1.

It is the policy of Western Governors University (WGU) that student capstone projects should not be based on or include, without authorization, restricted information. Restricted information is any proprietary or classified information or material belonging to your employer or any other third party. You acknowledge that you will not use restricted information in your capstone project without obtaining the third party's permission by using the "**IT Capstone Project Restricted Information Authorization Form**" found in the Supporting Documents section of Task 1.

DEGREE EMPHASIS: BSCSIA

ANALYSIS:

Project Topic

"Secure Campus Messaging at Scale: An Open-Source, Commodity-Based Edge Design"

Problem Statement or Project Purpose

K-12 schools rely on reliable information and are tasked with distributing it to their stakeholders (students, educators, staff members, parents, caretakers, and other community members). They use many methods of electronic communication to deliver information to their stakeholders, including email lists, social media groups, newsletter platforms, 3rd-party apps, and more. It is paramount that communications from trusted platforms are available, as they provide important and sometimes crucial information for students and caregivers to manage their students' educational pathways and experiences. If each student is to do their best in their educational experience, they need to be provided the opportunity to take part in every program offered by the institution, as it fits the students' needs and interests.

Also, K-12 schools commonly lack access to modern technology and solutions. These environments are often over-tasked with waves of activities to fulfill a school-year calendar, and these activities are spread across various waves, leaving many education technology concerns unaddressed when the time to develop a solution and/or the budget to complete a project is available. In a well-researched published report titled "Challenges and solutions when using

technologies in the classroom"[1], researchers from Arizona State University found that schools do not get access to education technology solutions due (in part) to three primary constraints: access to equipment and implementation, training on how to use the technology, and support barriers to technology integration (Johnson et al., 2016). Solutions that are not cost-effective, easy to deploy and manage, and do not fit the K-12 environment often leave gaps in the technology a school needs to operate effectively.

To support this concept, this paper proposes the design of an electronic messaging system to deliver secure displayboard functionality using well-known open-source packages and protocols, paired with low-cost commodity hardware, to create a reliable, secure system for use in the K-12 school context. As reflected in "Free and Open Source Software in Education", the author describes why open-source technology is effective in education contexts and states: "... a significant advantage to FOSS educational software is that it typically runs on less-powerful computer hardware than proprietary software, which reduces or even eliminates the need for cash-strapped school districts to upgrade their existing equipment, so those resources can be spent elsewhere."[2] The design system is intended to support the best of open technology, paired with commodity hardware (likely to be found in K-12 settings), to deliver commercial-like displayboard technology at very low cost, so that resources can be utilized elsewhere. With this solution, more schools would have access to better technology for communication purposes, aiming to present more opportunities to students and increase overall participation in their educational pathways. The paper will analyze the use cases and problems that an open-source, commodity-based secure messaging platform can deliver, and detail a working, lightweight prototype to complete the overall capstone effort.

DESIGN and DEVELOPMENT:

Project Scope

a. Project Goal(s) and Supporting Objectives

Goal: Deliver an end-to-end solution promptly (within 6 weeks) to prove secure solutions are possible with free and low-cost technologies to solve issues that may occur in resource-constrained environments, such as K-12 institutions and agencies.

Objective: Prove that a secure, enterprise-grade messaging delivery solution can be created using well-known open-source software and techniques, paired with low-cost hardware.

[1] Johnson, A. M., Jacovina, M. E., Russell, D. G., & Soto, C. M. (2016). Challenges and solutions when using technologies in the classroom. In S. A. Crossley & D. S. McNamara (Eds.), Adaptive educational technologies for literacy instruction (pp. 13–29). Taylor & Francis. https://files.eric.ed.gov/fulltext/ED577147.pdf

[2] Donnelly, M. (2021). Free and open source software in education. EBSCO Research Starters. https://www.ebsco.com/research-starters/computer-science/free-and-open-source-software-education

b. Project Outcomes and Deliverables

Outcome: Deliver a secure solution using free and low-cost technology for an institutional application

Deliverables:

- One capstone paper with use case analysis, solution design, and outcome write-up.
- Source code repositories and prototype code bases for the middleware and Raspberry Pi solution, along with technical documentation
- A demonstration video of the solution working end-to-end, with author-narrated technical design and guidance on reuse in a real environment.

IMPLEMENTATION and EVALUATION:

Describe how you will approach the execution of your project –

- Research an end-to-end solution, including technologies for secure message delivery, message signing and encryption, network protocols, packaging and distribution, and other concepts to deliver as envisioned.
- Write the first draft based on research and design the overall solution and approach. Revise until approved by Task 2 efforts.
- Implement a solution based on the design and prove end-to-end. Document what works and what does not work. Focus on proven, secure existing technology solutions, such as Postgres, Keycloak, Python libraries, and so on, to leverage what software engineering communities have found successful and to accelerate the project's work on the problem domain.
- Artificial Intelligence will be utilized for source code generation for the solution (e.g., Python, Node.js) and also as a research assistant in the technical aspects of the project. For example, do we need only singing or full encryption? What version of TLS is most secure and cost-effective for this solution to transfer encrypted information across the network? (weigh pros vs cons and trade-offs for each). Also, for code generation: generate a Python 3 console app that receives messages from a queue, uses a private key as primary authentication, and provides extensive, configurable logging so we can monitor its operation.
- Run the complete solution end-to-end and document the results. Create a demonstration video of the solution in action in a practical lab setting. Add the final write-up as Task 3.

IRB REVIEW:

- ☐ **√ - This project does not involve human subjects research and is exempt from WGU IRB review.**

COURSE INSTRUCTOR SIGNATURE: Jeff Davis

COURSE INSTRUCTOR APPROVAL DATE: 01/20/2026